



Next Generation 2
**CONVEYOR
MONITORING
SYSTEM**

SERIES 9403MLSS / 9404MLSS / 9406MLSS

MONITORING LUBRICATORS
INSTALLATION & PARTS MANUAL

V2.24/2.25



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System subject to change. Exact configuration may differ depending on options quoted/purchased.

Introduction



Mighty Lube Systematic Lubrication, Inc. with decades of field engineering and service experience, has developed a chain-wear monitoring system with a completely automatic lubrication system to serve a wide range of applications. Mighty Lube's concept of

monitoring chain wear is to reduce the need to replace conveyor chains by pinpointing failing chain links. In turn, centralized lubrication eliminates wasted man hours that would be spent manually greasing conveyor bearings, oiling chains, and filling individual lubrication tanks on individual conveyors.

Mighty Lube's Patented Conveyor Monitoring and Lubrication System monitors and records vital data that is not observable to the human eye. Real time data of Chain Speed, Drive Amps, Tank Level and Take-Up Pressure or distance, as well as daily chain wear readings shown both as numerical data and graphs. All the important data from all of your conveyors, plant wide can be monitored from your computer. Get instant, real time information vital to predictive and preventive maintenance without ever leaving your desk!

Mighty Lube can supply you with industry-tested lubricant for any given application. These lubricants are used on many types of conveyors, chains, and bearings; from high heat ovens to ambient temperature transfer lines and different types of washer conveyors, all with excellent results.

LIMITED WARRANTY

Mighty Lube Systematic Lubrication, Inc. warrants its lubrication equipment against defects in material and/or workmanship for a period of one year, from the date of sale, provided that:

The defect has not been caused by the purchaser or others. The defect is not the result of normal wear and tear. The defect is not the result of tampering or damage to the equipment. The defect was not caused by electrical or environmental conditions exceeding the standard operating conditions and environment. The equipment has been properly maintained by the purchaser. The equipment has been used for its intended and designed purpose.

Mighty Lube, at its option, will repair or replace the defective equipment or part(s) provided that the purchaser returns the defective equipment or part(s) to the Mighty Lube facility at purchasers' expense, or, purchaser provides written authorization and agreement to pay Mighty Lube's reasonable travel and labor expenses. This warranty applies only to the repair/replacement of parts, not shipping or labor to replace those parts.

This warranty shall not apply to any equipment or part(s) that have been repaired or replaced by the purchaser or another third party without Mighty Lube's consent and written authorization.

Mighty Lube will not be liable for any expenses incurred by the purchaser due to defect or failure of the Mighty Lube equipment or part(s). Mighty Lube's sole and total liability shall be the repair and/or replacement of the defective equipment or part(s). Mighty Lube makes no other warranties of any kind, expressed or implied.

Furthermore, with regards to Mighty Lube equipment, this warranty shall apply for five years from the date of purchase if Mighty Lube lubricants are used exclusively. In the event that Mighty Lube Lubricants are not used exclusively, then only lubricants approved and authorized by Mighty Lube shall be used and a standard one year would then be applied. Unapproved lubricants will void warranty coverage.

Mighty Lube makes no other warranties of any kind, expressed or implied. Mighty Lube shall not be liable for any consequential, incidental, special, punitive or other indirect damages, and Mighty Lube's total liability arising at any time from the sale or use of the equipment shall not exceed the purchase price paid for the equipment.



INSTALLATION

PROCEDURE 1: INSTALLATION OF THE POWER CONTROL UNIT AND PUMPING RESERVOIR

LOCATION

1. To select the proper location for the 7110DM/7165DM power control unit and pumping reservoir, place it in a convenient spot where it can be supplied with a constant 120 VAC power source. The location must be out of the main lift truck traffic area and where the reservoir will be accessible for filling. (Next to a building column or in a corner is recommended.) Nearest possible location of plant network Ethernet drop should be considered at this point, maximum distance via Cat 5e cable is 200 feet.

POWER SUPPLY

1. After the location is selected, the power control unit and pumping reservoir can be supplied from a constant 120 VAC standard US power outlet or may be wired direct, if so desired.

FILLING THE RESERVOIR

1. Fill the central pumping reservoir with lubricant. DO NOT plug in or turn on until PROCEDURE 6 – START-UP has been completed.

WARNING:

Mighty Lube equipment is designed to operate on Mighty Lube thin film penetrating lubricants of 60 S.U.S. or less at 100 F.





INSTALLATION

PROCEDURE 2: DATA COLLECTION UNIT LOCATION



LOCATION

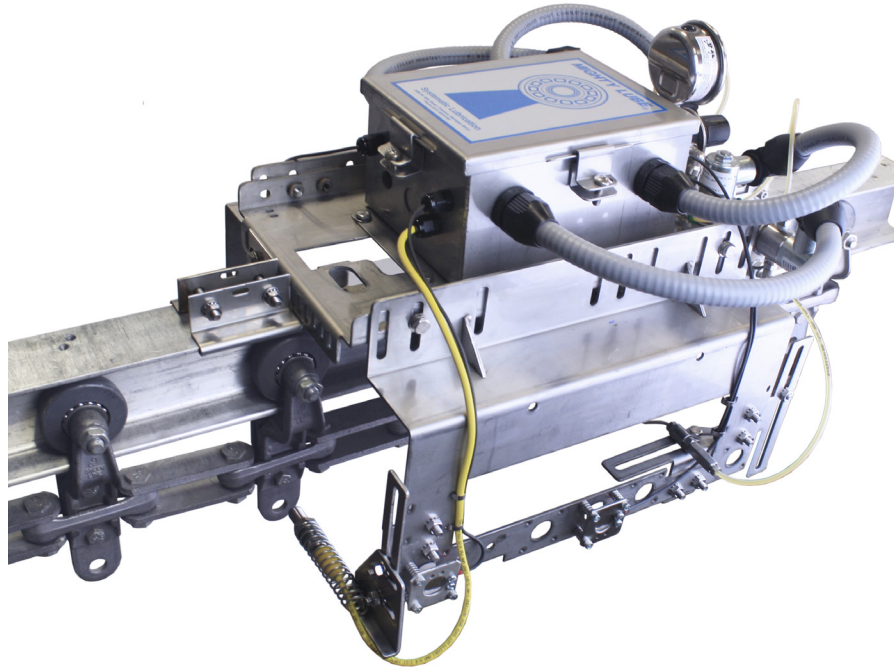
5200M Data Collection Unit should be placed in a dry location, away from excess dust and/or vibration. 5200M requires 120VAC and an Ethernet connection to operate. Ethernet connection must be on the same network as the 7110DM/7165DM Reservoir/Power Supply and the computer with Mighty Lube software installed. If more than one 7110DM/7165DM is to be installed, verify IP address before placement, as improper connection will effect system performance. If PC program is being updated, remove any earlier versions of the Mighty Lube software prior to installation.

Please see PC user guide for computer program installation and operation.



INSTALLATION

PROCEDURE 3: INSTALLATION OF THE MONITORING LUBRICATOR(S)



LOCATION

1. Monitors should be placed before the drive(s) near a conveyor take up. Not only does this ease the installation, it also makes future maintenance less complicated.

NOTE: Try to locate the lubricator in a place where routine service can be easily accomplished.

IMPORTANT: Conveyor chain monitor MUST have constant chain speed and tension.

MOUNTING

1. Select a straight section of the conveyor track or I-beam, approximately four feet long and free of obstruction.
2. Measure the lubricator's track or I-beam rail and remove the same size section of track or I-beam from the conveyor. (It may be necessary to disconnect the chain and move the chain and trolley out of the way).
3. After the track or I-beam rail is cut out and removed, set the lubricator in place, ***making sure the direction of travel is correct.***



INSTALLATION

4. Weld the lubricator track or I-beam in place.

NOTE: Do not weld across the bottom flange of the track or I-beam where the bearings will have to roll across the weld.

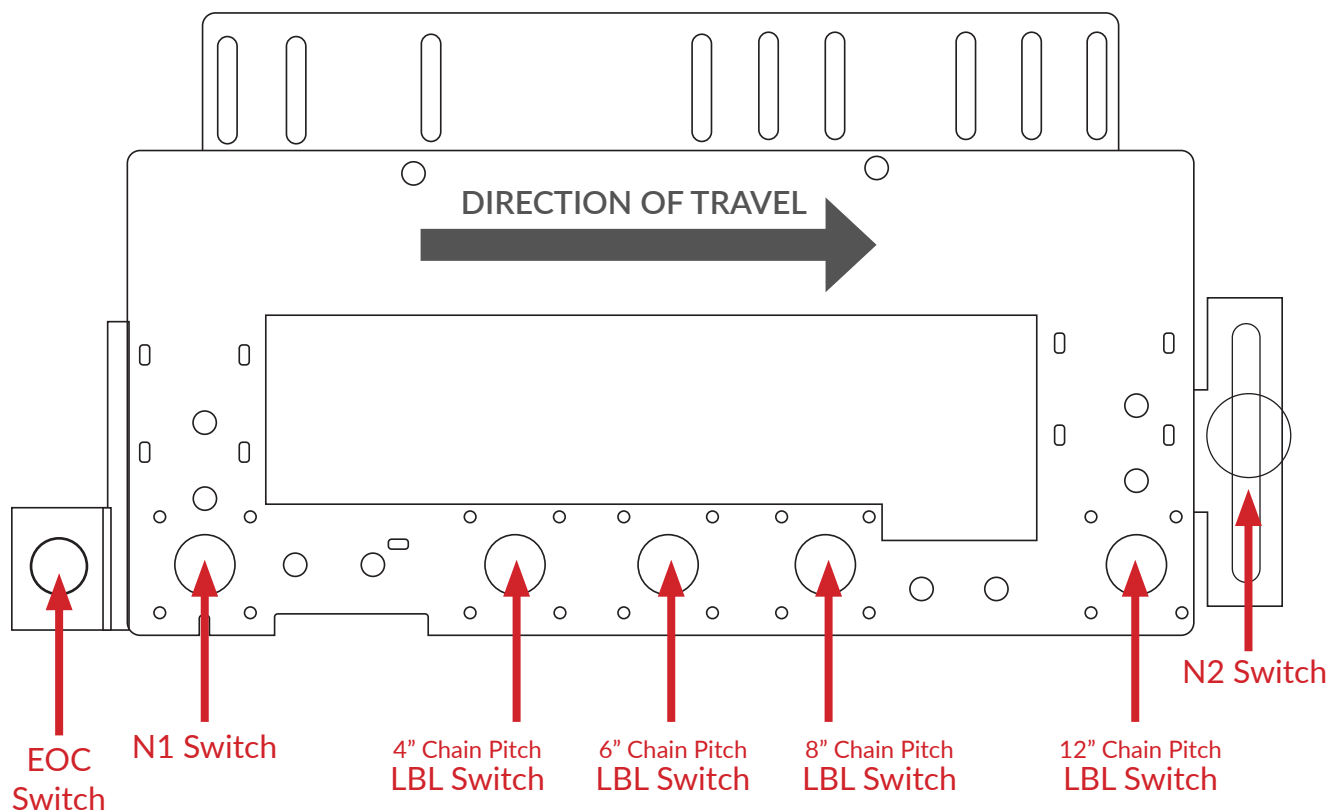
5. Once the lubricator is welded into place, re-connect the chain if disconnected.

FINAL ADJUSTMENT

1. Adjust the N1 (Link) switch until it is activated by the center of the chain link. N1 emitter switch is mounted to a spring-loaded bracket. This bracket is attached to the side plates with three 8-32 screws. Tightening upper screws will move switch down. Tightening left screws to move switch to the right and so on. **Refer to drawing showing horizontal switches. Figure 1 (4" I-Beam Shown for Reference)**

***NOTE: Switches are factory set and do not usually require adjustment.**

FIGURE 1



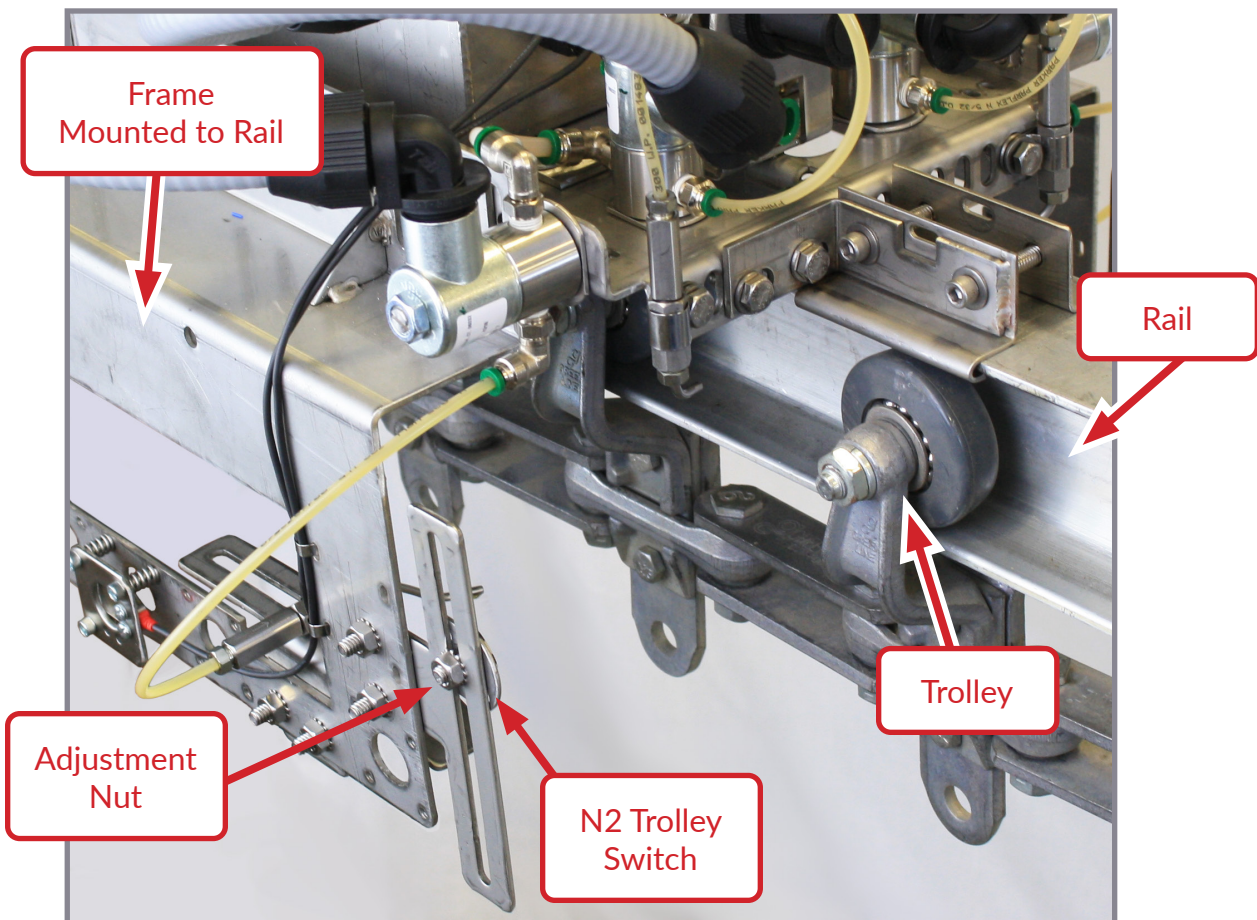


INSTALLATION

FINAL ADJUSTMENT (Continued)

2. Adjust the N2 (Trolley) switch until it is activated by the yoke or arm of the load bearings. **Refer to the drawing showing horizontal switch. Figure 2**
3. Adjust Link By Link (LBL) switch by moving it to the appropriate hole according to your specific chain pitch. **Refer to the drawing showing horizontal switch. Figure 1**

FIGURE 2





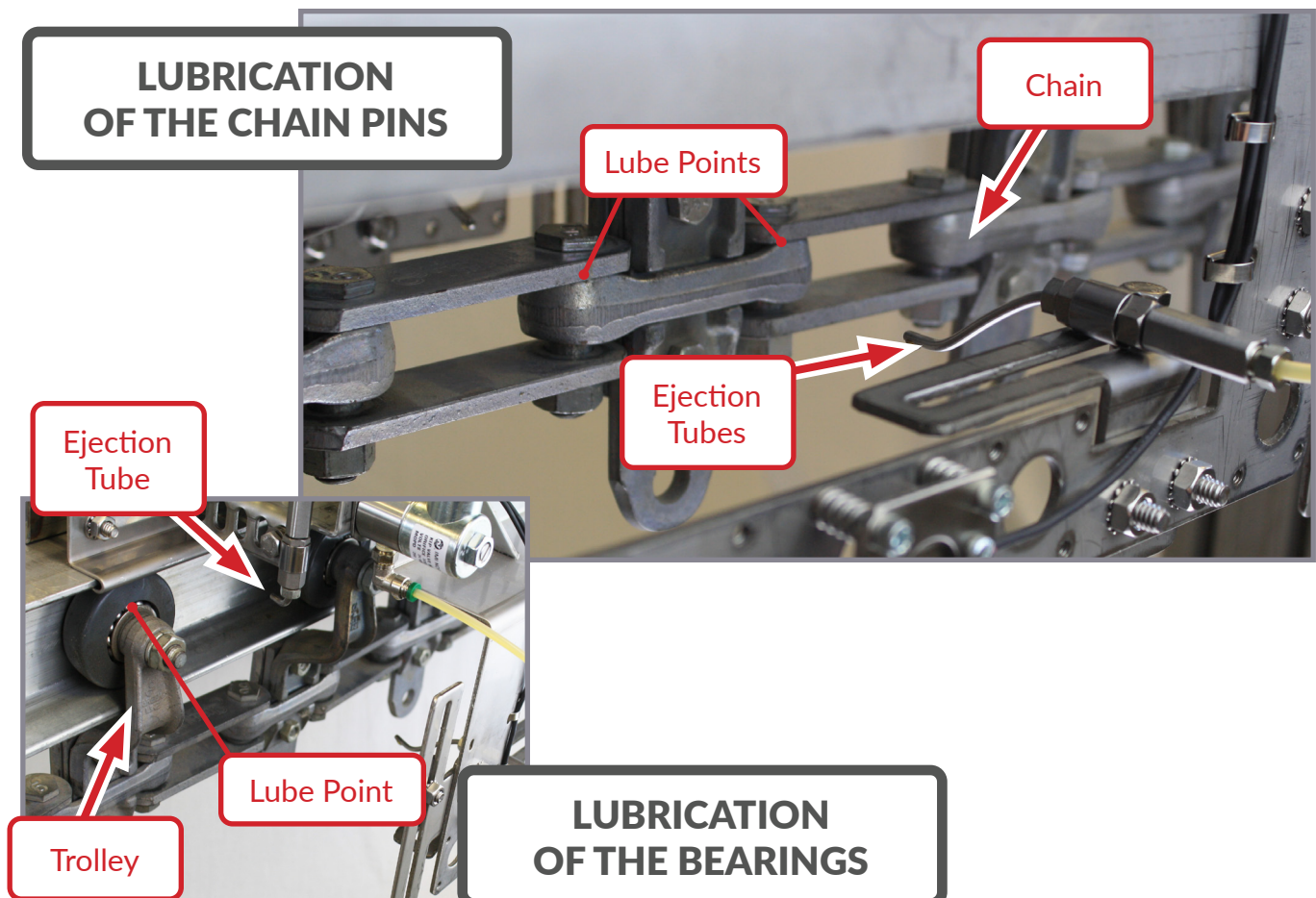
INSTALLATION

FINAL ADJUSTMENT (Continued)

4. Adjust LBL switch until it is activated by the center of the chain link. The LBL switch is mounted to a spring loaded bracket. This adjustment bracket is attached to the side plates with three 8-32 screws. Tighten top screws to move the switch down. Tighten left hand screw to move switch to right and so on.

NOTE: The ejection tubes are pre-set; however, minor adjustments may be required. The lubricant should be applied to the chain pivots and to the balls of the trolley bearings. See figure 3

FIGURE 3



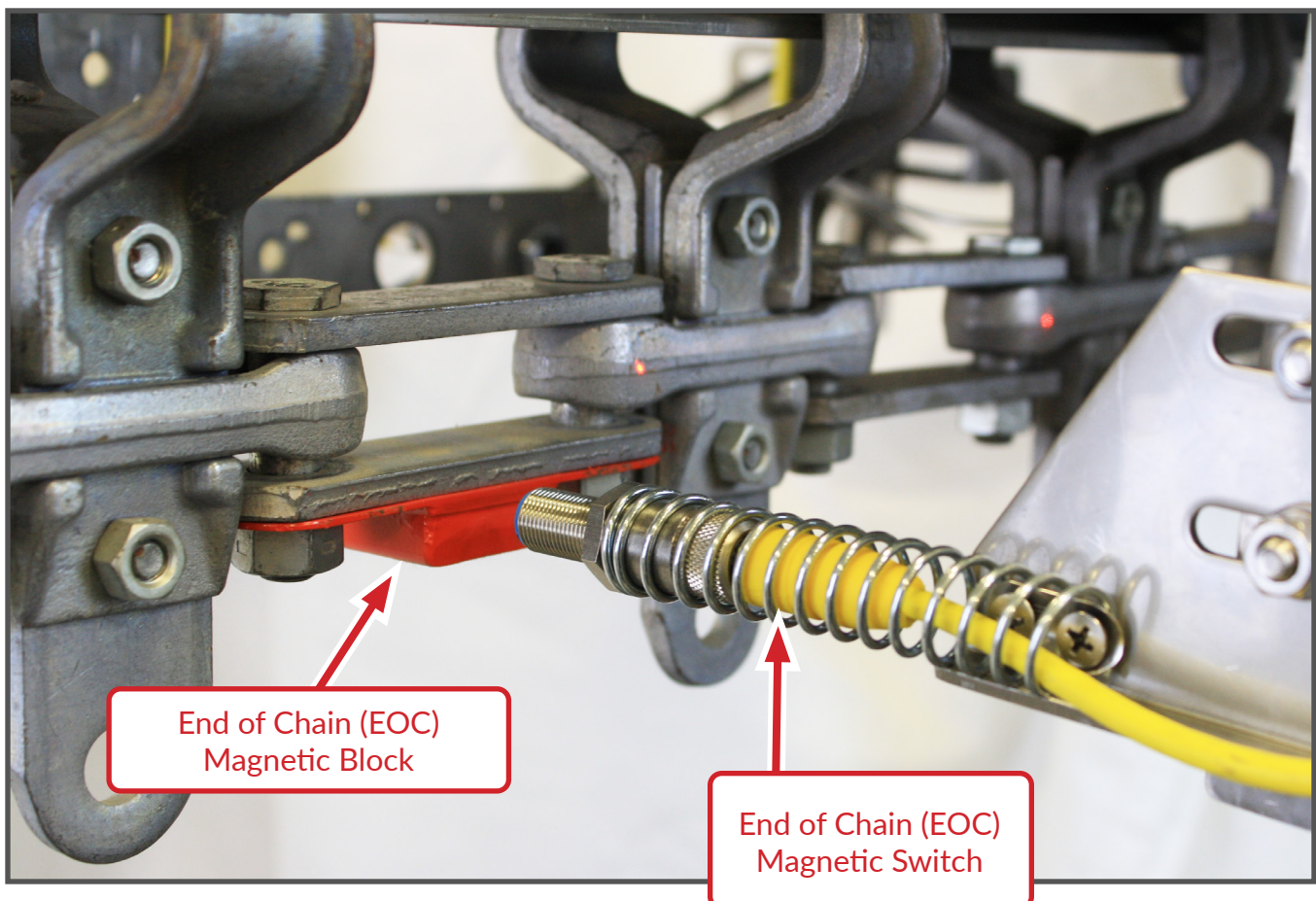


INSTALLATION

FINAL ADJUSTMENT (Continued)

5. Install End of Chain (EOC) magnetic block with the magnet facing the EOC Switch and adjust the magnetic pickup switch. Pay special attention to the relation to the chain to prevent entanglement. EOC Switch should be adjusted to approximately 1/2" of the EOC Block as mounted to chain. [Refer to Figure 4](#)

FIGURE 4



WARNING: Incorrect adjustment will prevent system function.



INSTALLATION

PROCEDURE 4: INSTALLATION OF LUBRICANT SUPPLY LINE(S) AND CONTROL CABLE(S)

GENERAL INFORMATION

1. Lubricant supply lines are normally $\frac{3}{8}$ " O.D. nylon tubing. Steel or copper tubing may also be used where high temperatures or custom requirements are a factor. Rubber or poly tubing should be avoided.
2. Power/LAN Cables are usually 4 conductor. 18 gauge twisted pair cable, RED/BLACK and WHITE/BLACK. One pair RED/BLACK and one pair WHITE/BLACK.
See wiring diagram page 28.
3. Control cables are normally 12-conductor, 18 gauge wire with RED, BLACK, GREEN, GREY, VIOLET, LIGHT BROWN, PINK, YELLOW, BROWN, BLUE, ORANGE and WHITE wire colors. *See wiring diagram page 28.*
4. Mighty Lube equipment is designed to operate on thin film penetrating lubricants of 60 SUS or less at 100 degrees Fahrenheit.
NOTE: Specialty lubricants may require specialty equipment for proper function.

CONNECTION

Lubricant supply line must be connected from the central pumping reservoir to the first lubricator. Subsequent lubricators and equipment are connected in a series *as shown in Figure 5.*

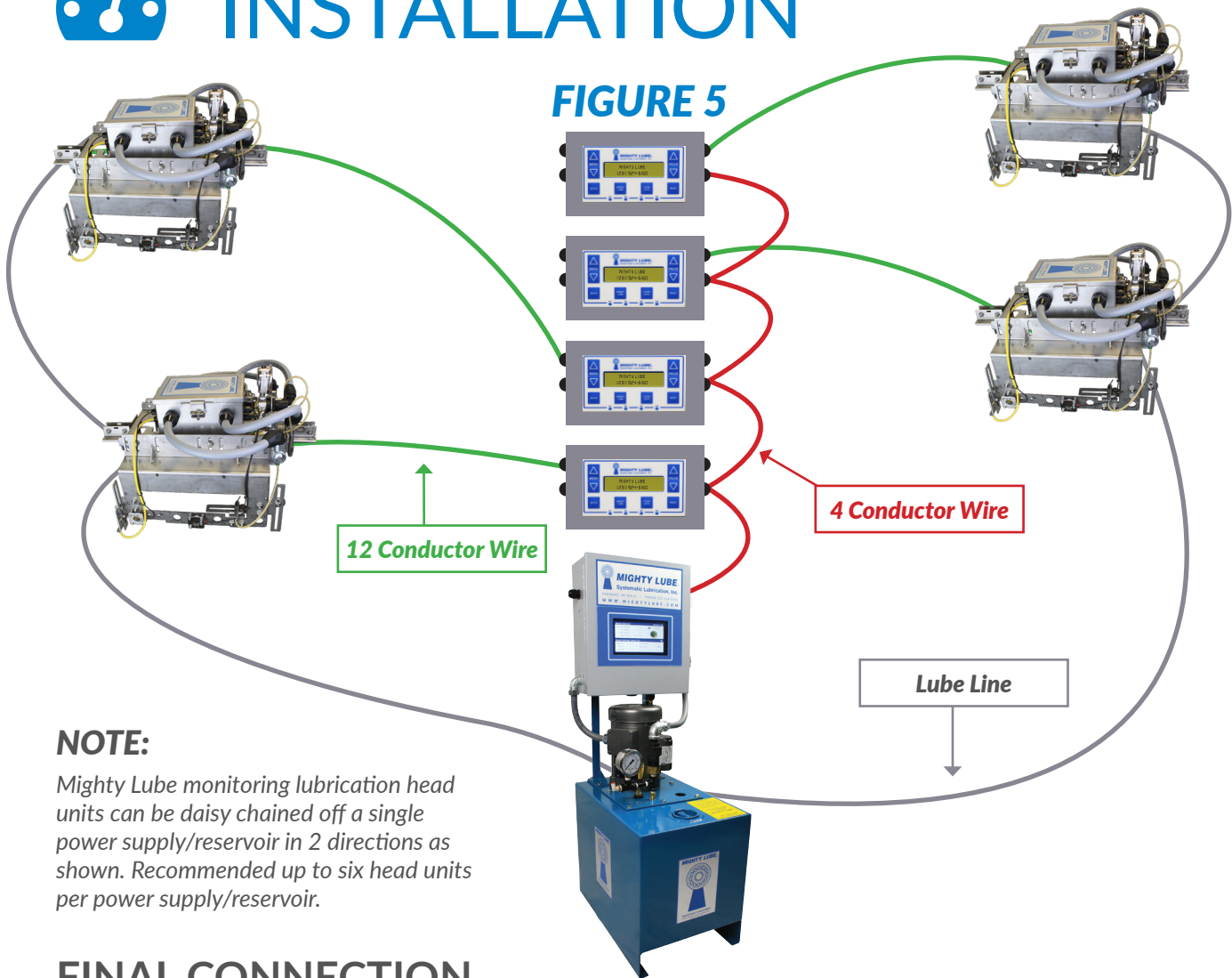
Lubricant supply line(s) and control cables(s) are normally fastened overhead along building structural components by means of tie wraps and beam clamps. When installing the lubricant supply line(s), always tape the end closed to prevent contamination in the tubing.

NOTE: Leave approximately one foot of extra tubing and wire at each lubricator and approximately three to four feet extra at the power control unit and pumping reservoir for the final connections.



INSTALLATION

FIGURE 5



NOTE:

Mighty Lube monitoring lubrication head units can be daisy chained off a single power supply/reservoir in 2 directions as shown. Recommended up to six head units per power supply/reservoir.

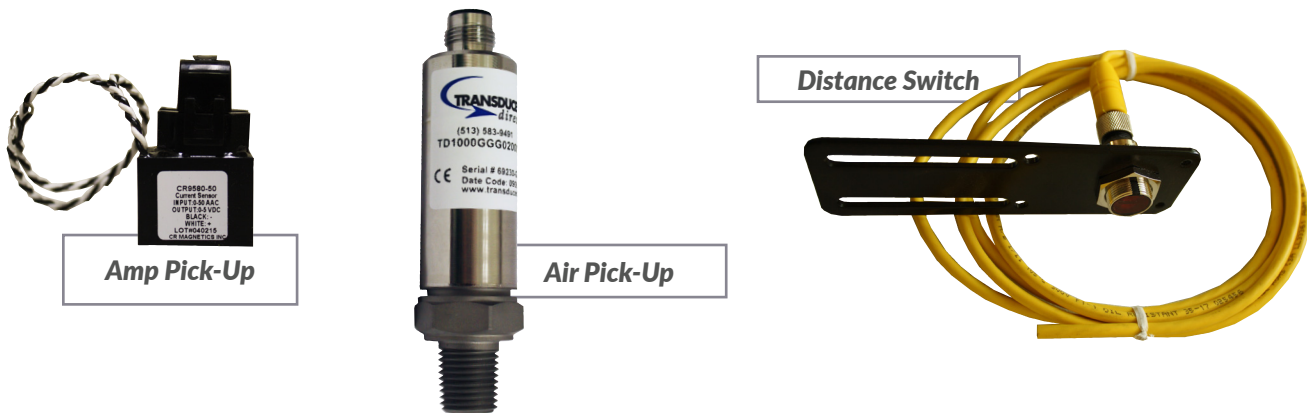
FINAL CONNECTION

1. Attach the lubricant supply line by cutting it to the proper length (see note above) and pushing it as far as possible into the fitting on the lubricator. If necessary, tighten the nut on the fitting.
2. Attach the power to the lubricator **See wiring diagram page 28**. Observe the color coding.
3. Attach the lubricant supply line to the pumping reservoir by pushing it as far as possible into the fitting. If necessary, tighten the nut on the fitting.
4. Attach the power/LAN to the power control unit **as shown in wiring diagram page 28**. Observe the color coding.



INSTALLATION

PROCEDURE 5: AMP PICK-UP, AIR PICK-UP AND DISTANCE SWITCH INSTALLATION



The complexity of this installation can be minimized by use of the provided wire and junction box(s). The installation is commonly broken into steps, and are described as follows.

See wiring diagram page 28.

1. From the main control (MLR-72) box run cable to drives and the take-ups. 2 conductor cable is required for each drive and 3 conductor cable is required for the take-up. (3 conductor cable is required for DC drives)
2. The amp pick-up(s) must be installed on the power wires feeding the drive motor. The pick-up is installed on only one of the wires to the drive. Take care to observe polarity on the DC amp pick-ups.
3. The air pick-up(s) are installed on the regulated air feed line to the cylinder(s). It is very important that the air pick-up have the same pressure as is applied to the take-up cylinder. Commonly the feed line is spliced and a ball valve is installed just prior to the air pick-up, this allows future maintenance.
4. The distance switch(s) must be placed to allow detection of conveyor take-up extension/retraction. Often the distance switch will be placed above the counterweight to detect downward travel. Use bolts or beam clamps to secure distance switch bracket to take-up framework.



INSTALLATION

5. The distance switch and/or air pick-up requires power from the control board to operate. So observe color coding when connecting inside the junction box and in the lubricator control box. [See wiring diagram page 28.](#)
6. Some applications require the use of distance switch instead of or in conjunction with an air pick-up.

NOTE: Confirm all equipment and wiring will not interfere with operation of conveyor.

If you have any questions please contact your local authorized sales and service representative.

PROCEDURE 6: START-UP

1. Make sure all lubricant supply line(s), power/LAN cables and control cable(s) are connected at the monitoring lubricator(s), power control unit, and pumping reservoir. Fill the pumping reservoir if not yet filled.
2. Turn on the unit.

WARNING: Mighty Lube equipment is designed to operate on Mighty Lube thin film penetrating lubricants of 60 S.U.S. or less at 100 F.



INSTALLATION

PROCEDURE 7: FINAL ADJUSTMENTS

PROGRAMMING THE CONTROL BOARD

The following instructions explain how to program the control board for proper operation of the monitor lubricator. Reference the diagram below for all programming steps. [Figure 6](#)

FIGURE 6





INSTALLATION

PROGRAMMING THE CONTROL BOARD (Continued)

KEYPAD DESCRIPTION		
MENU		Allows user to scroll display menu.
VALUE		Allows user to change program values.
SETUP		Prompts user to password screen and accesses initial setup functions. (Password is 6160)
DIRECT LUBE SWITCH		The Direct Lube Switch will cause the lubricator to lubricate up to 999 seconds. This allows checking of the ejection tube aim without beginning an entire lubrication cycle. It also can be used to check for proper switch/solenoid function.
CHAIN DATA		Gives chain wear data for previous revolution. Use the VALUE key to view specific link wear data by link and 10' wear.
RESET SWITCH		The Reset Switch is used to reset the control board. This will cause the lubricator to enter a lube cycle and begin counting bearings and links from zero. This will not reset Programmed N1 count when EOC (end of chain) function is active.



INSTALLATION

PROGRAMMING 9400M SERIES

(When entering numbers on the console, single clicks adjusts single numbers; holding the VALUE key adjusts numbers by 10s and 100s).

Refer to Figure 6 on page 13. Start at screen displaying Mighty Lube (231) 924-6160. Press the SETUP button to access password screen. Press and hold the UP VALUE key until you reach 6160. Next press SETUP to access initial setup functions which are listed below. Press menu down button to scroll to next menu setting. (These settings are factory set, inaccuracy may cause the system to fault.)

INITIAL SET-UP			
SET-UP MENU	MENU VALUES	FACTORY SETTING	DESCRIPTION
METRIC UNITS	Y or N	N	Allows user to select metric or standard units.
LUBE MENU	Y or N	Y	Enables lubrication portion of system.
WEAR SPD MIN	Numerical	05	Minimum Speed of chain wear monitoring.
SPD TIMEOUT	Numerical	30	Delay before alarm trigger after start-up.
CAT BACKUP	Y or N	N	Sets system drive function for Main/Backup or Slave Drives.
TIMED LUBE*	Y or N	N	Allows user to set specific time of day to lubricate.
MONITOR DOG	Y or N	N	Enables dog-spacing alarm function. Dog space monitoring disables Drive 4 input.
ENABLE N2 Error	Y or N	N	Enables missing trolley detection alarm function.
ENABLE EOC SW	Y or N	Y	Allows EOC counter reset.

*** Timed lube function will be disabled after 18 hours if time cannot be established.**



INSTALLATION

PROGRAMMING 9400M SERIES (Continued)

SET-UP MENU	MENU VALUES	FACTORY SETTING	DESCRIPTION
N1/EOC ERR	0-250	000	EOC error offset prevents Lubrication if EOC is not detected (EOC must be enabled). Factory setting does not prevent lubrication
PAINT MENU	Y or N	N	Paint Chain Menu
STATION ADDR	1-99		EACH CENTRAL SYSTEM MONITOR MUST HAVE A DIFFERENT STATION ADDRESS.
CHAIN PITCH	02.1 - 99.1	06"/08"/12"	Length of distance from the start to finish of chain link. SET AS NEEDED.
LBL ADJUST	Numerical	0.00	Add or remove up to 0.25" to each chain link.
DIR LUBE TM	1-999	60 seconds	Length of time solenoid valves will operate during test function.
EXT LUBE N1	Y or N	N	External Lube Control N1 (Chain Pin)
EXT LUBE N2	Y or N	N	External Lube Control N2 (Trolley)
EXT LUBE N3	Y or N	N	External Lube Control N3 (Carrier Trolley)
EXT LUBE N4	Y or N	N	External Lube Control N4 (Guide Wheels)
SPEED DELAY	Y or N	N	Allows for lubrication offset as conveyor speed changes.
ENABLE P CHG	Y or N	Y	Allows user to change P values.



INSTALLATION

PROGRAMMING 9400M SERIES (Continued)

SET-UP MENU	MENU VALUES	FACTORY SETTING	DESCRIPTION
SWITCH STATUS #1		N1 -- -- -- --	Displays which switches are activated. Switch name/number will be displayed on screen when active.
SWITCH STATUS #2		-- 10 LBL -- --	Displays which switches are activated. Switch name/number will be displayed on screen when active.
ANALOG INPUT #1-4	OFF/DIST	OFF	Enable Analog #1-4 input for distance sensing
ANALOG INPUT #5-16	OFF/AMPS/ AIR/TEMP/ VIBR		Configure Analog #5-16 input for specific conveyor variations.
OUTPUT TEST (toggle selection ON by pressing Direct Lube and change selected output by pressing value keys)	ON/OFF	N1: OFF S2: OFF S1: OFF CATB: OFF CATM: OFF N4: OFF N3: OFF N2: OFF	The output test confirms operation of the solenoid valves without interrupting the program.

PRESS SETUP AGAIN TO EXIT SETUP MENU AND SAVE CHANGES.



INSTALLATION

PROCEDURE 8: PROGRAMMING LUBRICATION

(From the start-up screen, press the menu down button to scroll to next function).

LUBRICATION MENU	DESCRIPTION
N1 - INPUT 1 (LINK NUMBER)	N1 number refers to the number of chain link sets to be counted in one complete revolution of the conveyor. This value is automatically entered into the program if the EOC (END OF CHAIN) magnetic block is properly installed on the chain with the magnet facing the switch and that the switch is properly adjusted to sense the block as it passes. The EOC function must be set to Y in the setup menu.
C1 - OUTPUT 1 (OFF-CYCLES)	Off-cycles for input 1 refers to the number of revolutions the conveyor will make between applications of lubricant. Scroll to C1 by pushing the menu down arrow. Using the value up/down switch, program the number of desired link off-cycles. For constant lubrication, enter 0000 or 0001. To lubricate every other revolution of the chain, enter 0002. For every fifth revolution, enter 0005 and so on. The number of cycles without lubrication plus one on-cycle.
T1 - INPUT 1 (LUBE TIME)	Lube time refers to how long the solenoid valve sprays lubricant onto the chain as it passes by the lubricator. Scroll to T1 by pushing the menu down arrow. Use the value up/down switch to program the duration in milliseconds, of the solenoid operation for each link that is being lubricated. Typical settings are about 0010 to 0015 for I-beam systems. It is recommended that this time be set no lower than 0007 to insure proper activation of the solenoid.
D1 - LINK SWITCH DELAY TIME	Delay time refers to how long the switch must wait before counting again. This feature is designed to prevent the valve from “double firing” and dispensing lubricant to an undesirable location. Scroll to D1 by pushing the menu down arrow. Use the value up/down switch to program the delay time for the link switch. However, if the time is too long, you may miss the next link. Link delay time is adjustable from 0.000 seconds (no delay) to 3.999 seconds (maximum delay).



INSTALLATION

LUBRICATION MENU	DESCRIPTION
N2 – INPUT 2 (TROLLEY OR BEARING NUMBER)	N2 number refers to the exact number of trolley bearing pairs/sets (I-beam conveyor) in one complete revolution of the conveyor. Scroll to N2 by pushing the menu down arrow. Using the value up/down switch, program the number of trolleys on the chain. If the N2 value is set at 0000, this cycle will automatically be updated along with the N1 (links). If the exact number is not known, see the following ALTERNATE METHOD OF PROGRAMMING N2 (TROLLEY NUMBERS).
ALTERNATE METHOD OF PROGRAMMING N2 (TROLLEY NUMBERS)	1. Program 9999 into N2 (trolley number) portion of the control board using the value switches. 2. Mark or place a tag on the conveyor directly in front of the lubricator. This allows you to keep track of the conveyor's progress. 3. Press the reset switch to bring counters to zero. 4. Start the conveyor and let it run one complete revolution until the mark or tag appears in the same place in front of the lubricator. Note the "C" (or Count) number in the display. This is the exact number to be programmed into the N2 (trolley number) portion of the control board. Use the value up/down switch to enter the desired number. 5. Programming N2 (trolley number) is now complete.
C2 – OUTPUT 2 OFF-CYCLES	Off-cycles refer to the number of revolutions the conveyor will make between applications of lubricant. Scroll to C2 by pushing the menu down arrow. Using the value up/down switch, program the number of desired trolley off-cycles. For constant lubrication, enter 0000 or 0001. To lubricate every other revolution of the chain, enter 0002. For every fifth revolution, enter 0005 and so on. Normally programmed to same number as C1.



INSTALLATION

LUBRICATION MENU	DESCRIPTION
T2 - OUTPUT 2 LUBE TIME	Lube time refers to how long the solenoid valve sprays lubricant onto the trolley bearings as they pass the lubricator. Scroll to T2 by pushing the menu down arrow. Use the value up/down switch to program the duration, in milliseconds, of solenoid operation for each trolley or vertical bearing that is being lubricated. Typical settings are about 0010 0015 for I-beam systems. It is recommended that this time be set no lower than 0007 to insure activation of the solenoid.
D2- INPUT 2 DELAY TIME	Delay time refers to how long the switch must wait before counting again. Scroll to D2 by pushing the menu down arrow. Use the value up/down switch to program the delay time for the trolley switch. This function prevents the switch from reactivating for the programmed number of milliseconds. This is useful for preventing double firing resulting in lubricant being dispensed in an undesired location, or the same location from being over lubricated. D2 delay time is adjustable from 0.000 seconds (no delay) to 3.999 seconds (maximum delay).
NC - CAT DRIVE NUMBER	Cat drive number refers to the number of engagement dogs on the cat drive chain. Scroll to NC by pushing the menu down arrow. Using the value up/down switch, program the number of Cat Drive drive dogs. <i>Note: The main cat drive solenoid valve is activated by the lubricator link switch. It does not require its own switch to operate. On units with a single drive, the cat drive lubricator need only be wired in accordance with the schematic found in the control box. If main and back-up drives are used, the lubricator with the switch must be installed on the back-up drive and wired according to the schematic. Refer to Wiring Diagrams on page 28. No additional programming is required if the back-up drive is used.</i>
CC - CAT DRIVE OFF-CYCLES	Off-cycles refer to the number of revolutions the cat drive chain makes between applications of lubricant. Scroll to CC by pushing the menu down arrow. Using the value up/down switch, program the number of desired off cycles. For constant lubrication, enter 0000 or 0001. To lubricate every other revolution of the chain, enter 0002. For every fifth revolution, enter 0005 and so on. A usual setting for a drive is 1000 to 3000 depending on the environment.



INSTALLATION

LUBRICATION MENU	DESCRIPTION
TC1-TC8 CAT DRIVE LUBE TIME	Lube time refers to how long the solenoid valve sprays lubricant onto the cat drive chain. Scroll to TC by pushing the menu down arrow on the menu screen. Use the value up/down switch to program the duration, in milliseconds, of solenoid operation for each dog and link that is being lubricated. It is recommended that this time be set no lower than 0007 to insure activation of the solenoid. <i>Refer to Wiring Diagrams on page 28.</i>
N3 – INPUT 3 AUXILIARY NUMBER	This function is useful for lubricating such things as free carriers or spinners and requires the use of an additional switch to sense the additional lubrication point(s). For this auxiliary example we will use a free carrier system. Auxiliary number refers to the number of free carriers to be counted in a power and free conveyor system. Scroll to N3 by pushing the menu down arrow. Using the value input switches, program the number of auxiliary bearings.
C3 – INPUT 3 OFF-CYCLES	Off-cycles refer to the number of times all of the free carrier will pass the lubricator between applications of lubricant. Scroll to C3 by pushing the menu down arrow on the menu screen. Using the value up/down switch, program the number of desired off cycles. For constant lubrication, input 0000 or 0001. To lubricate every other revolution of the chain or cycle of the free carriers, enter 0002. For every fifth revolution or cycle, enter 0005 and so on.
T3 – OUTPUT 3 LUBE TIME	Lube time refers to how long the auxiliary valve will spray lubricant onto the bearings and guide rollers of the free carrier as it passes the lubricator. Scroll to T3 by pushing the menu down arrow. Use the value up/down switch to program the duration, in milliseconds, of solenoid operation for each point that is being lubricated. It is recommended that this time be set no lower than 0007 to insure activation of the solenoid.



INSTALLATION

LUBRICATION MENU	DESCRIPTION
D3 – INPUT 3 SWITCH DELAY TIME	Delay time refers to how long the switch must wait before counting again. This function prevents the switch from reactivating for the programmed number of milliseconds. This is useful for preventing double firing resulting in lubricant being dispensed in an undesired location or the same location from being over lubricated. Scroll to D3 by pushing the menu down arrow. Use the value up/down switch to program the delay time for the link switch.
N4 – INPUT 4 AUXILIARY*	This function is useful for lubricating such things as free carriers or spinners and requires the use of an additional switch to sense the additional lubrication point(s). For this auxiliary example we will use a free carrier system. Auxiliary number refers to the number of free carriers to be counted in a power and free conveyor system. Scroll to N4 by pushing the menu down arrow. Using the value input switches, program the number of auxiliary bearings.
C4 – INPUT 4 OFF-CYCLES*	Off-cycles refer to the number of times all of the free carrier will pass the lubricator between applications of lubricant. Scroll to C4 by pushing the menu down arrow. Using the value up/down switch, program the number of desired off cycles. For constant lubrication, input 0000 or 0001. To lubricate every other revolution of the chain or cycle of the free carriers, enter 0002. For every fifth revolution or cycle, enter 0005 and so on.
T4 – OUTPUT 4 LUBE TIME*	Lube time refers to how long the auxiliary valve will spray lubricant onto the bearings and guide rollers of the free carrier as it passes the lubricator. Scroll to T4 by pushing the menu down arrow. Use the value up/down switch to program the duration, in milliseconds, of solenoid operation for each point that is being lubricated. It is recommended that this time be set no lower than 0007 to insure activation of the solenoid.
D4 – INPUT 4 DELAY TIME*	Delay time refers to how long the switch must wait before counting again. This function prevents the switch from reactivating for the programmed number of milliseconds. This is useful for preventing double firing resulting in lubricant being dispensed in an undesired location or the same location from being over lubricated. Scroll to D4 by pushing the menu down arrow. Use the value up/down switch to program the delay time for the link switch.

*** Options will not be available if N2 error is enabled in set-up menu.**



INSTALLATION

TIME OF DAY OPTIONS

Time of day lubrication featured are only available when TOD is enabled in Set-Up Menu.

See page 15.

MENU	SETTING	DESCRIPTION
LT1	P __:__	Preset lube “window” start time 1, 24 hr format
LT2	P __:__	Preset lube “window” start time 2, 24 hr format
LT3	P __:__	Preset lube “window” start time 3, 24 hr format
LT4	P __:__	Preset lube “window” start time 4, 24 hr format
LD Minutes	P0000	Lube “window” duration. Minutes allowed for lubrication,

** If TOD is enabled, lubrication becomes cycle end time based. Lubrication will only occur if off cycles have elapsed and TOD indicates lubrication should begin.*

PAINT MARKER OPTIONS

These will only be available if paint marker is enabled in set-up menu. *See page 15.*

LUBRICATION MENU	SETTING	DESCRIPTION
PAINT WEAR	P0.00	Amount of chain wear in link, value equal or greater will be marked.
PAINT TIME	P0000	Length of paint spray time (recommended 0100ms)
PAINT DIST	P016	Distance from N1 to paint nozzle, minimum distance is 2 chain pitches
PAINT ON	N/Y	Enable paint marker for one revolution of the conveyor.



INSTALLATION

LUBRICATION MENU	DESCRIPTION
CHAIN FT/M	Describes the speed of chain in feet per minute. Based on the programmed chain pitch over 3 switch inputs.
10' LN0000	Displays wear of the last 10' section of chain measured.
LBL LN0000	Displays wear of the last link measured.
A01-A16	Displays analog input information, varies depending on monitor setup.
MAIN HRS	Number of hours that the main drive has been operating since last reset or install.
BKUP HRS	Number of hours that the backup drive has been operating since last reset or install.
DOG	Reports link number of incorrectly spaced pusher dog.
N200 N400 ERRO	Reports missing or bent trolleys when enabled
COM ADDR	Displays station address and current communications status. Each monitor head must have a different station address number. The station address can be changed in SETUP menu.
Clock __:__:__	Displays local time as set by DCU Lubrication is disabled if DCU connection is unavailable for 18 hours.



PARTS

Parts List: Head Units 9403MLSS, 9404MLSS, 9406MLSS (MONITOR & LUBRICATOR)

INDEX NUM.	PART NUMBER	DESCRIPTION
1	MLR-72	Complete Electronic Monitor Control Enclosure
*	MLR-70MO	Microprocessor Board MO
2	MLR-12	Liquid Heyco Fitting
*	MLR-71	9200 Enclosure With Membrane Switch
*	MLR-1SS	1/8" Pipe to 1/4" Tube Fitting Elbow (Stainless Steel)
*	MLR-3SS	1/8" Pipe to 1/4" Tube Tee (Stainless Steel)
4	MLR-2SS	Solenoid Valve (Stainless Steel)
5	MLR-66	Ejection Nozzle / Nut / Ferrule
6B	MLR-88R	Laser Switch (receiver)
6A	MLR-88E	Laser Switch (emitter)
7	MLR-13SS	System #2 Plate - 3" (Stainless Steel)
8	MLR-14	#6-32 Machine Screw
9	MLR-22	Hex Head Machine Screw, Lock washer, Nut (Stainless Steel)
11	MLR-9SS	Stainless Steel Ty-wrap
12	MLR-34	End of Chain Magnetic Switch
13	MLR-74SS	End of Chain Spring Bracket Assembly (Stainless Steel)
*	MLR-36	End of Chain Magnetic Block
*	MLR-38SS	U Bracket (Stainless Steel)
*	MLR-39SS	Clamp Bracket (Stainless Steel)
*	MLR-40	Air Transducer
*	MLR-42	AMP Pickup
*	MLR-53	Distance Switch

* Not Shown



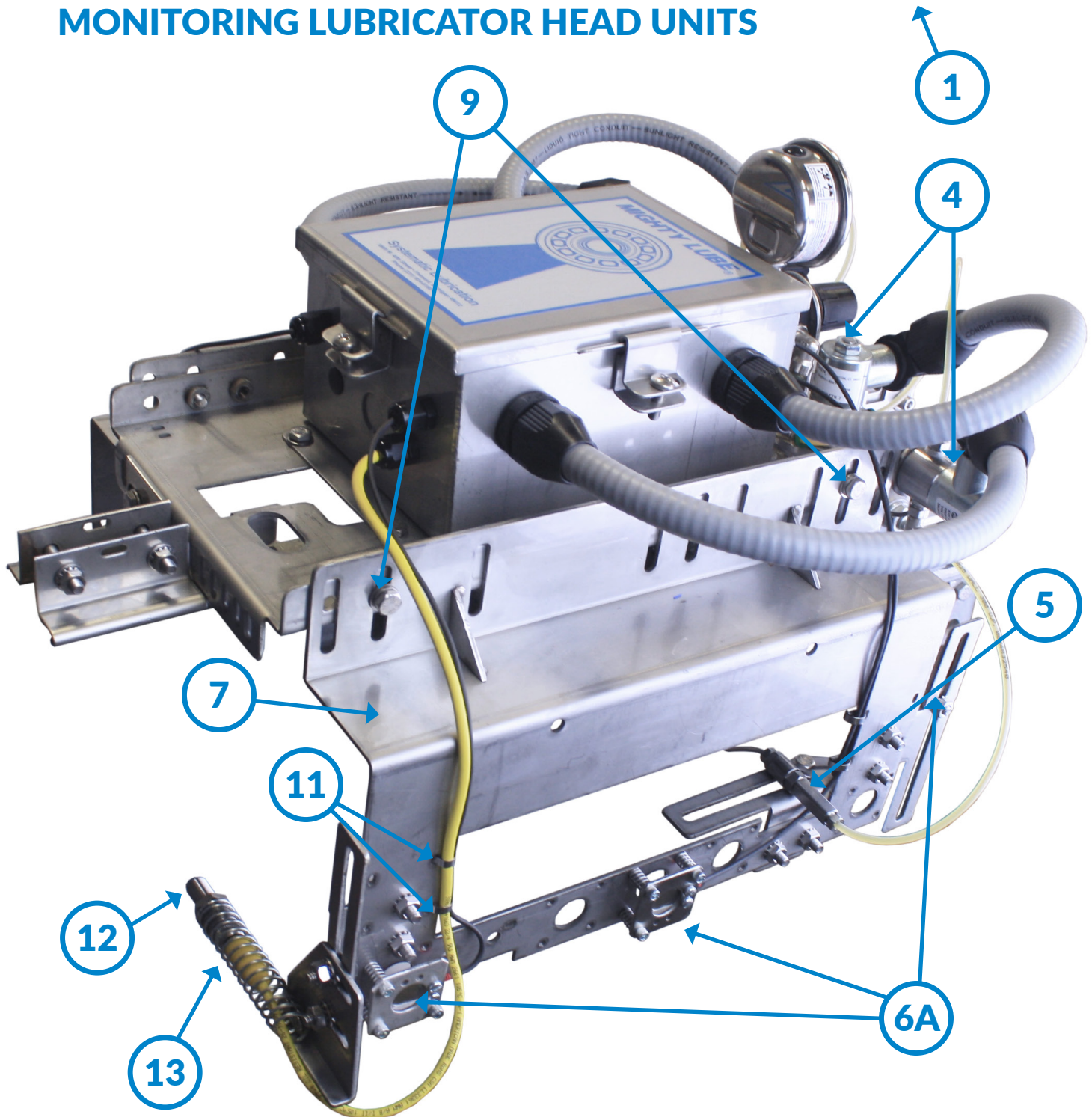
PARTS

Diagram: Front View

9403MLSS, 9404MLSS, 9406MLSS

MONITORING LUBRICATOR HEAD UNITS

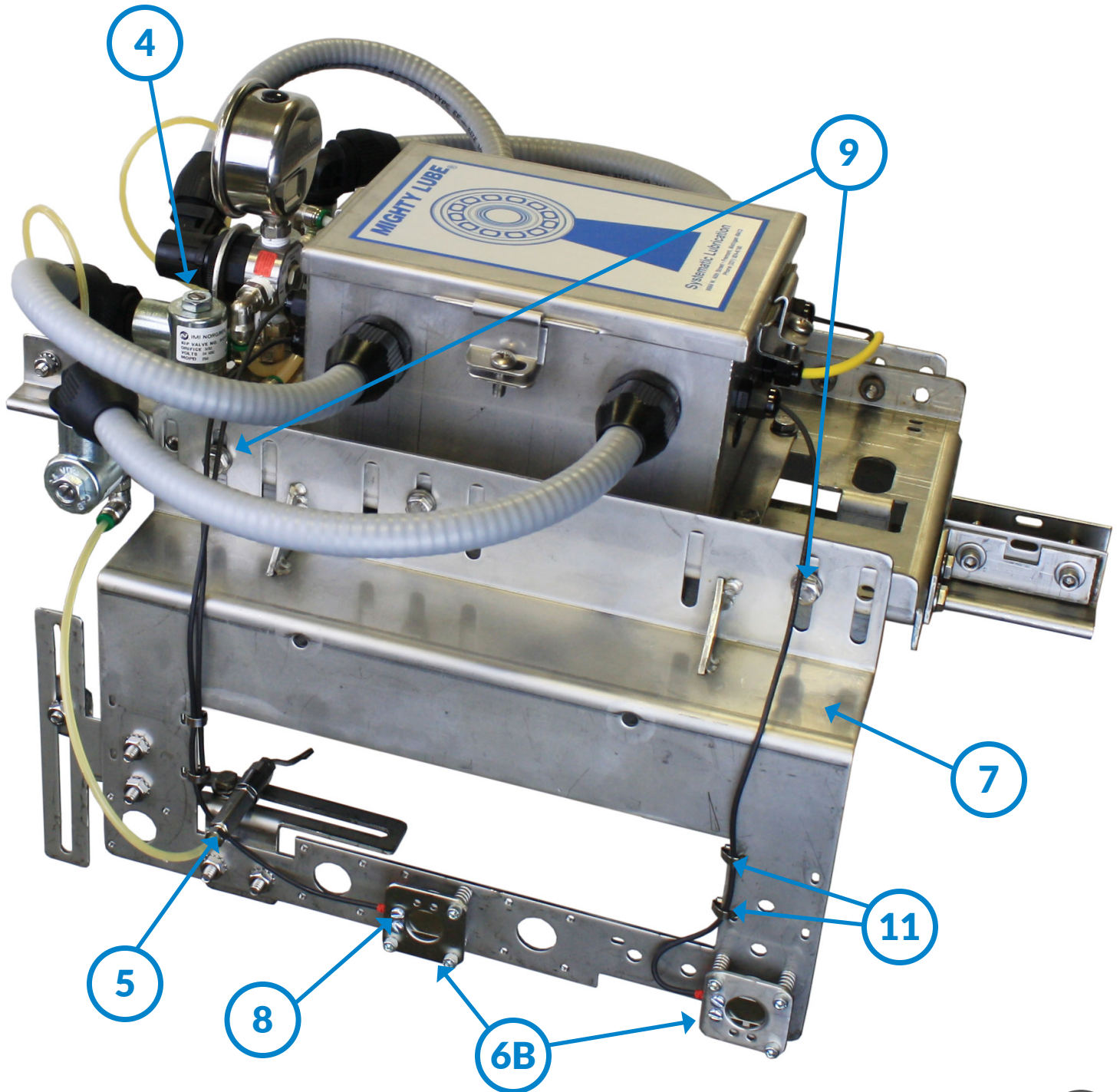
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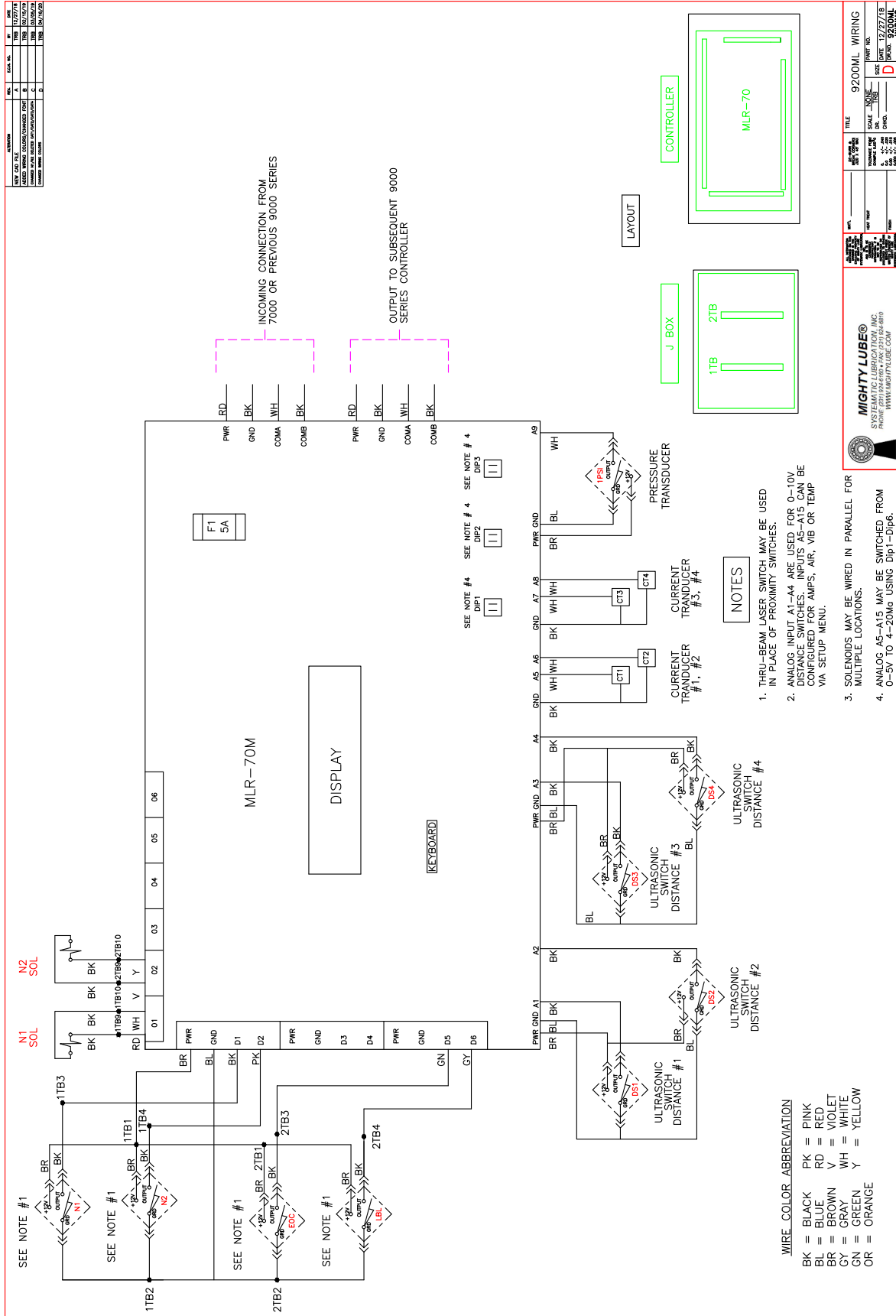
PARTS

Diagram: Back View 9403MLSS, 9404MLSS, 9406MLSS MONITORING LUBRICATOR HEAD UNITS





WIRING DIAGRAM



MIGHTY LUBE®
SYSTEMATIC LUBRICATION
PHONE (771) 524-2160 • FAX (771) 524-4690
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9200ML WIRING

SCALE: 1/8" = 1'-0"

DATE: 12/27/18

REV: D

QUESTIONS?

**CONTACT
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